

# Studying Attribute Pruning and Prediction Performance Trade-offs in Face Attractiveness Prediction

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*Dataset links:* [CelebA](#), [Facial Beauty Rating](#)

## Abstract

This project looks at how reducing the number of input features via pruning affects the performance of machine learning models. We use the CelebA dataset, which consists of facial images paired with annotated attributes. Beyond the standard tabular setup, we bring in low-level image features — pixel values, histogram features, gradient-based features, and RGB statistics — and combine them with the original tabular attributes to form a single unified feature representation to predict the attractiveness of a person. We train three models — Logistic Regression, Random Forest, and Gradient Boosting — and evaluate how each one holds up under different levels of feature pruning. Pruning is applied systematically to reduce the number of input features, and each model is retrained at every pruning level so we can track how performance shifts. To check whether our findings are generalizable, we run the same pipeline on a second dataset, SCUT-FBP5500, and compare the trends across both datasets. The aim of this study is to understand the trade-off between cutting down features and maintaining predictive performance, and to see how different models behave when the input they rely on is reduced via aggressive pruning.

## 1 Introduction and Contributions

In machine learning, we often see research papers trying to improve the prediction accuracy by increasing the number of features. However, this hurts model performance when irrelevant or redundant features are introduced to the feature space [2]. This common phenomenon is called the *curse of dimensionality*, where, as the number of features increases, the feature space increases in sparsity, which can often lead to overfitting, reduced generalizability, and increased computational cost [2]. This is a common scenario in high-dimensional settings, like combining two input types - tabular and image features. Pruning, a method that removes the input features of low importance with the goal of reducing the dimensionality of feature space, has become a popular technique to improve efficiency as well as scalability. Beyond this, pruning has also been shown to help improve the generalization performance of machine learning models [1]. However, pruning without proper analysis can impact model performance. This paper aims to study how pruning affects predictive performances on high-dimensional feature spaces when input data includes binary tabular attributes and image features. We specifically focus on face attractiveness as it allows us to combine the two different input types and focus on the impact of feature reduction rather than maximizing accuracy, which many papers focus on. We explore how much pruning is too much by looking at the predictive accuracy across various models and datasets. Specifically, we focus on studying the impact of feature reduction on the performance of three different models (Logistic Regression, Random Forest, and Gradient Boosting), across two datasets (CelebA and SCUT-FBP5500), and using seven pruning levels ranging from 0% to 90%.

The main contributions of this paper include:

- A systematic evaluation of pruning across different pruning levels, where each of the models is retrained after each pruning level to ensure a fair comparison.
- We build a unified feature representation that combines tabular attributes and low-level image features into a single feature vector.

- Moreover, we include a comparison of three model families, which are Logistic Regression, Random Forest, and Gradient Boosting, and analyze how each model type performs under pruning.
- We validate our findings across two datasets that are of different sizes and use different label types - CelebA and SCUT-FBP5500. Here, we observe how our pruning pipelines generalize across the two datasets.
- Finally, we provide evidence that, on average, pruning up to 75% offers minimal loss of performance across the two datasets, suggesting that feature reduction can be helpful without sacrificing predictive accuracy.

## 2 Related Work

Feature selection and representation learning have both been widely studied in the context of high-dimensional data, but they are usually approached from the perspective of improving predictive accuracy rather than understanding trade-offs in feature reduction. For example, Xu et al. [4] propose a gradient boosted feature selection method that integrates feature selection directly into the learning process. Their approach captures non-linear relationships between features while remaining scalable, and it does well compared to other feature selection techniques by focusing on identifying a compact and informative subset of features [4]. Liu et al. [3] take a different angle and focus on learning strong feature representations from images, showing that combining low-level visual patterns with higher-level semantic information leads to improved attribute prediction performance [3]. Their work also shows that deep networks can automatically learn meaningful concepts like facial attributes through large-scale pre-training, which highlights how important rich feature representations are. Yang et al. [5] study supervised feature selection specifically in the context of high-dimensional image-derived feature spaces, showing that a lower-dimensional feature subspace can be identified that maintains high decision performance with little degradation in computational and modeling costs. While this provides evidence that aggressive dimensionality reduction need not collapse performance, their work focuses on finding the best possible compact subset rather than systematically studying how performance shifts across multiple levels of reduction. All three of these papers are mostly concerned with making features better or more informative, and none of them directly examine what happens to model performance as the feature space is progressively reduced. That is what this project is about – looking at how models behave under different levels of feature pruning, especially when working with combined tabular and image features, to better understand the trade-off between reducing dimensionality and maintaining performance of models.

## 3 Background Knowledge

Feature pruning is a technique used to reduce the number of input features by removing those that are of lower importance or redundant. This helps simplify the model and can improve efficiency, especially when working with high-dimensional data. Instead of training on all available features, the model is trained on a subset of features selected based on some numeric metric. Different types of models handle feature reduction differently. Linear models like Logistic Regression rely directly on the input features and can be more sensitive to noise or loss of important features. In contrast, ensemble methods such as Random Forest and Gradient Boosting are generally more robust, since they can capture non-linear relationships and distribute importance across multiple features. Understanding how feature reduction impacts model performance is critical, especially in tasks involving complex data representations like facial attributes, where both low-level image features and high-level semantic attributes contribute to prediction.

## 4 Problem Description

The primary goal of this work is to study how aggressive feature pruning impacts predictive performance in high-dimensional datasets. Specifically, we aim to understand how much feature reduction can be applied before model performance begins to degrade significantly. Starting with a combined feature space that includes both tabular attributes and image-based features, we apply varying levels of pruning — from

no pruning to highly aggressive pruning — and evaluate how different models respond to these changes. Furthermore, to assess the generalizability of these findings, the same pruning strategy is applied across two distinct datasets: CelebA and SCUT-FBP5500. By comparing results across these datasets, the objective is to determine whether pruning behavior and performance trends remain consistent across different data distributions, feature types, and labeling schemes.

## 5 Proposed Method

We begin with the CelebA dataset, which contains tabular features representing human-defined attributes such as smiling, hair type, and makeup. In addition to these attributes, we extract low-level image features directly from the images, including pixel values, histogram features, and gradient-based features. Each image is resized to  $16 \times 16$ , and from this representation, we extract a total of 825 image features: 768 flattened pixel values, 6 channel statistics (RGB mean and standard deviation), 48 histogram features, and 3 gradient-based features. These image features are combined with the 39 tabular attributes to form a unified feature vector of 864 features per image.

This setup spans both high-level semantic attributes and low-level visual information, creating a high-dimensional feature space that gives us a meaningful context in which to study how model performance changes as input features are reduced through pruning. We train three models on this feature space:

- **Logistic Regression** — serves as a linear baseline and is also used for feature ranking.
- **Random Forest** — included for its robustness to high-dimensional feature spaces.
- **Gradient Boosting** — used to capture more complex non-linear relationships.

## 6 Experiments

We evaluate all three models — Logistic Regression, Random Forest, and Gradient Boosting — under multiple levels of feature pruning. The pruning levels include no pruning at 0%, light pruning at 10%, 20%, and 30%, moderate pruning at 50%, and aggressive pruning at 75% and 90%. At each pruning level, features are ranked based on importance, and only the top features are retained. All models are retrained from scratch after pruning at each level to ensure a fair comparison.

Model performance is evaluated using accuracy and F1-score. Accuracy provides an overall measure of correctness, while F1-score captures the balance between precision and recall, giving a more complete view of model performance.

On the CelebA dataset, we first compare performance across three feature configurations — tabular-only, image-only, and combined — using both accuracy and F1-score. This allows us to determine whether incorporating low-level image features alongside tabular attributes provides additional value over using tabular data alone.

To validate that our findings are not limited to a single dataset, we apply the same pruning-based evaluation to the Facial Beauty Rating dataset (SCUT-FBP5500). For this dataset, we directly use the combined feature representation (tabular + images) and evaluate all three models across all pruning levels, using accuracy and F1-score, to examine whether similar pruning behavior is observed.

## 7 Results and Conclusion

After comparing the image-only, tabular-only, and combined datasets, and looking at both accuracy and F1-score, we see that the combined dataset performs better overall. This suggests that adding low-level image features does provide additional value beyond just the tabular attributes. Based on this, our further analysis focuses on the combined dataset, where we study how pruning affects model performance across all three models.

## CelebA Dataset

Looking at how the models behave under pruning on the CelebA dataset, Logistic Regression remains stable from 0% to 50% (with accuracy levels between 0.805–0.806), then starts to decrease at the 75% mark to 0.803. The real drop can be observed at the 90% pruning level, where accuracy drops to 0.797. However, for the F1-score, it remains notably stable at both the 75% and 90% pruning levels, holding at 0.809 and 0.808 respectively, suggesting that the model continues to maintain a consistent balance between precision and recall even as accuracy declines under aggressive pruning (see Figure 3).

Across all three models, performance remains fairly stable during light pruning (10%, 20%, and 30%) and well past moderate pruning (50%). The main changes start to appear around the 90% pruning level. Random Forest depicts a stable upward pattern (test accuracy: 0.787 at 0% to 0.799 at 90%, F1: 0.791 at 0% to 0.807 at 90%), while Gradient Boosting follows a relatively flat pattern throughout (test accuracy: 0.799–0.801, F1: 0.805–0.804). Logistic Regression performs the best in accuracy at most pruning levels and is the only model that shows a meaningful decline to 0.797 at 90% pruning. Overall, Random Forest and Gradient Boosting do not drop as sharply with pruning and remain relatively consistent even as the number of features is reduced (See Figure 1 & Figure 3).

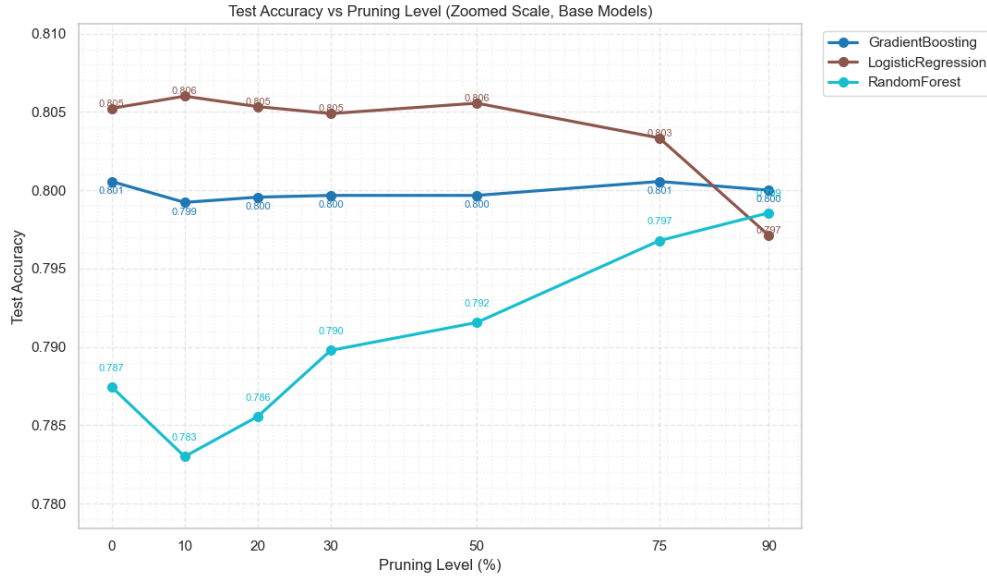


Figure 1: Test accuracy vs pruning level on CelebA across all three models

## SCUT-FBP5500 Dataset

On the SCUT-FBP5500 dataset, results were expected to vary compared to CelebA due to its facial beauty ratings being more subjective and continuous, which introduce label noise when converted into a binary classification task. When comparing the tree models, we saw that the results were stable under various ratio of pruning (see Figure 2):

- **Logistic Regression** performs best at 30% pruning (test accuracy: 0.754, F1: 0.744), suggesting that light pruning may help reduce noise in the feature space.
- **Random Forest** shows an interesting pattern, where performance dips slightly at light pruning levels (10–20%), then steadily improves from the 30% ratio onward before reaching its peak at 75% pruning (test accuracy: 0.739, F1: 0.727), and then decreases again at 90%.
- **Gradient Boosting** is the most stable out of all the models across all pruning levels. It performs best at 20% pruning (test accuracy: 0.734, F1: 0.708) and maintains accuracy around 0.727–0.734 throughout the graph.

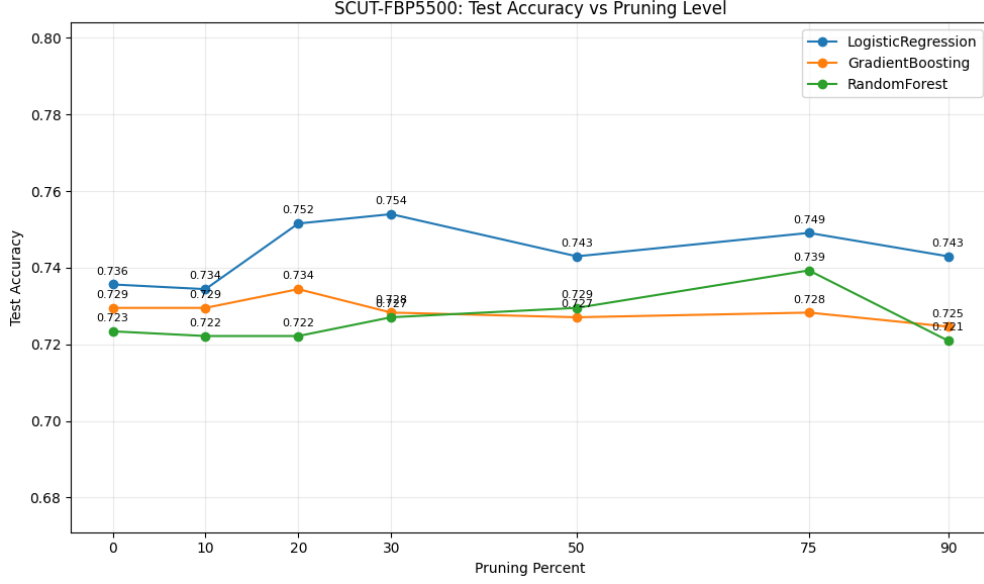


Figure 2: Test accuracy vs pruning level on SCUT across all three models

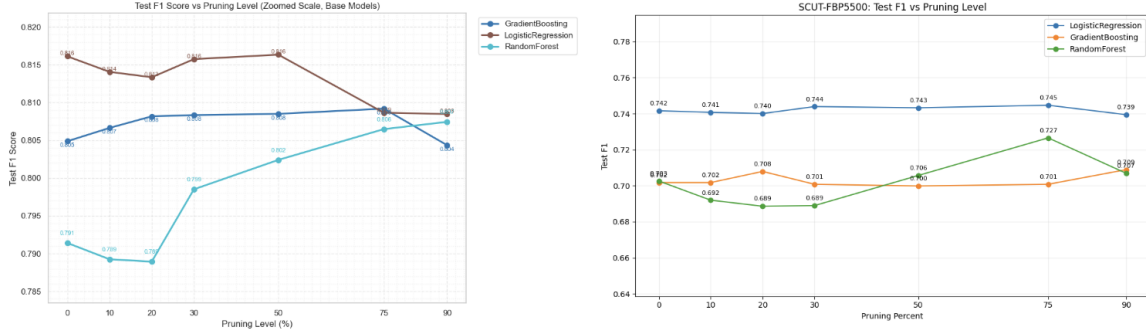


Figure 3: F1 Score for CelebA (left) and SCUT (right)

### Cross-Dataset Comparison

In CelebA, accuracy levels group around within 2% of each other across the three models at various pruning levels, indicating that the strong performance was supported by the large and rich combined feature space. Results vary more in SCUT, where Logistic Regression leads (peaking at 30% pruning), while Gradient Boosting and Random Forest trail behind at most pruning levels, suggesting that simpler models generalize better on smaller or noisier datasets.

Both datasets depict an overall similar trend: light pruning (10–30%) does not cause much change, and none of the models completely collapse under aggressive pruning. In CelebA, even at the highest pruning level (90%), the lowest observed test accuracy is 0.797 (Logistic Regression at 90%), while SCUT shows 0.721 (Random Forest at 90%) — both remaining useful. The consistent pattern that we observed in the two different datasets that varied in their sizes, feature types, and labeling properties suggests that the pruning trends that were observed are consistent across datasets.

Overall, since we are working with both image features and tabular attributes, Logistic Regression is useful for interpretability and feature ranking but is more sensitive to aggressive pruning, while Random Forest and Gradient Boosting are more robust to feature reduction. In summary, **on average across all the three**

**models and two datasets, pruning up to the 75% mark results in minimal loss in performance,** making it a feasible threshold for reducing feature dimensionality while preserving the core information needed by the models.

## 8 Main Challenges

One of the main challenges in this project was combining two very different types of data: tabular features and image-based features. The tabular data consisted of binary attributes such as smiling, glasses, and other facial characteristics, while the image features were continuous values like pixel intensities, histograms, and gradient-based features. Since these do not naturally fit together, we converted each image into a numerical representation and appended these as additional columns to the tabular dataset, resulting in a unified feature vector of 39 tabular features and 825 image features per image.

Another challenge was computational cost. Extracting features from a large number of images and retraining models across multiple pruning levels was time-intensive. To address this, we resized images to  $16 \times 16$  and relied on straightforward feature extraction methods — pixel values, histograms, and gradients — rather than more complex approaches. That said, resizing introduced some information loss, as fine details like facial expressions and texture were reduced. We treated this as a deliberate trade-off, including multiple feature types (histogram features for color distribution and gradient features for edge and structural information) to retain as much useful signal as possible.

Finally, the combined dataset resulted in high dimensionality, with 864 features per image, which can lead to poor generalization where the model starts to memorize noise instead of learning meaningful patterns. This makes the learning problem harder, since the model has to process a large number of features, many of which may be redundant or less informative. To handle this, we applied feature pruning at multiple levels (0%, 10%, 20%, 30%, 50%, 75%, and 90%), retraining models at each level using only the most important features. This helped reduce noise, improve efficiency, and allowed us to examine how much information the models actually need to perform well.

## 9 Team Contributions

| Member           | Contributions                                                                                                                                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aranya Venugopal | Data preprocessing and feature engineering for CelebA dataset (202K+ images); built unified feature vectors (tabular + image features); implemented Logistic Regression as baseline model.                                     |
| Sachi Vyas       | Implemented Gradient Boosting model for non-linear learning; designed and executed feature pruning pipeline; evaluated model performance using accuracy and F1-score and identified key insights along with visuals.           |
| Shazia Muckram   | Implemented Random Forest experiments and cross-model pruning analysis; integrated SCUT-FBP5500 (score-to-binary labels, pipeline setup, and pruning experiments); built core visuals and summarized CelebA vs. SCUT insights. |

## References

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